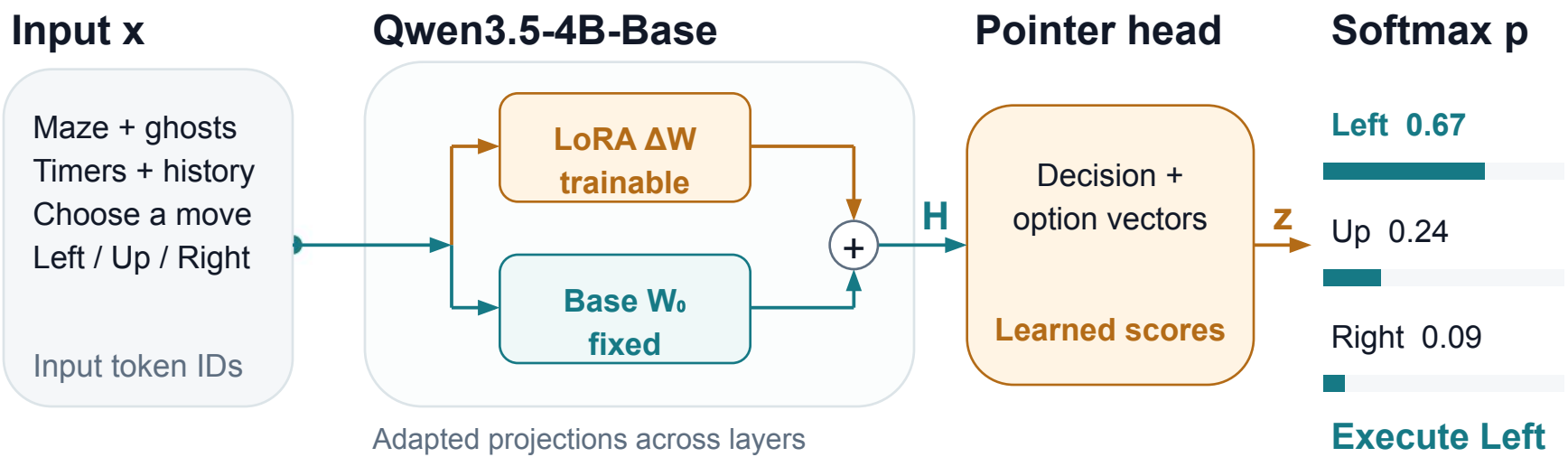
# Kev architecture

One board, supplied moves, direct probabilities




$$H = f_\theta(x)$$

$$p = \mathrm{softmax}(z)$$

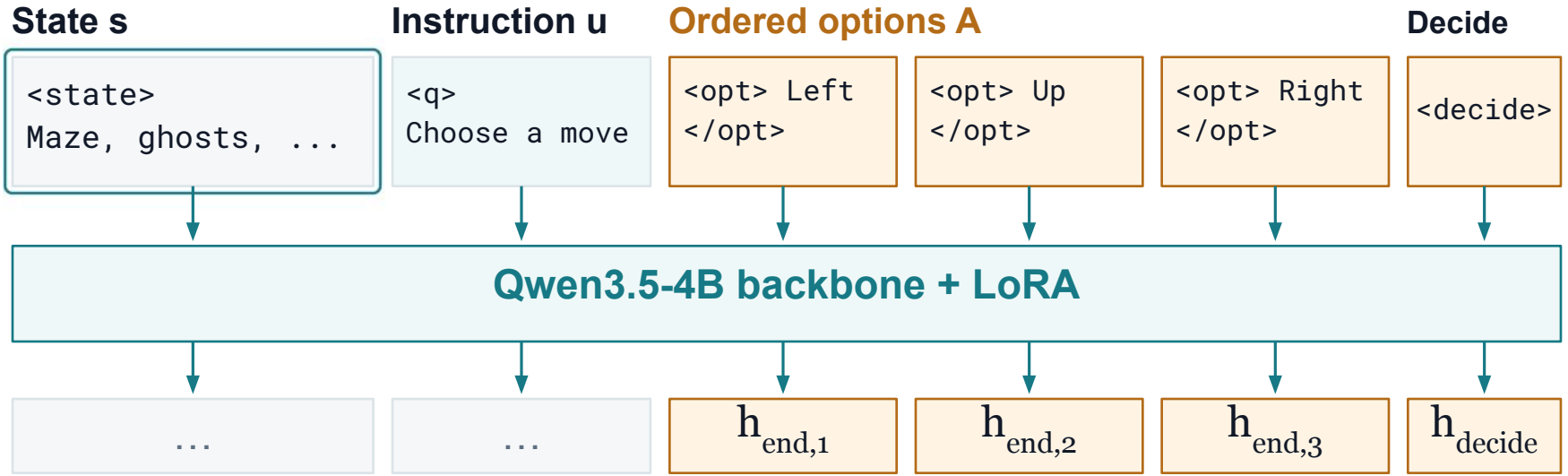
x: input tokens  H: hidden vectors  θ: effective backbone parameters

$W_0$: fixed base weights  ΔW: LoRA updates  z: scores  p: probabilities

Illustrative probabilities. Open animation controls.

# From board state to hidden vectors

State + instruction + legal moves → one structured input




The pointer head reads each option ending and the final decision vector

$$x = \text{encode}(s, u, A)$$

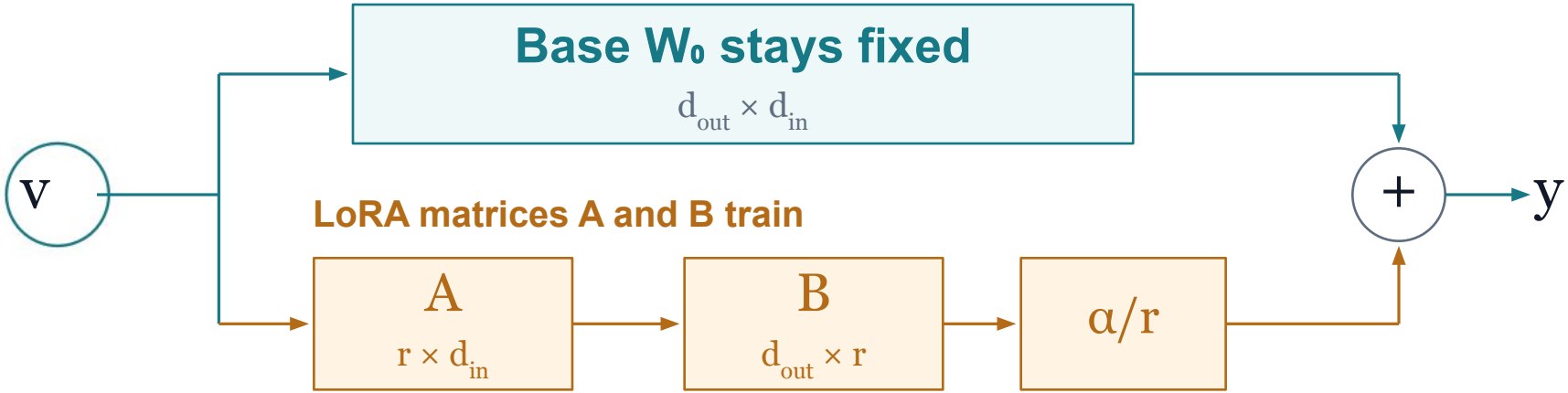
$$H \in \mathbb{R}^{L \times d}$$

L tokens × d features

Schematic token groups. Readable marker names. Open animation controls.

# LoRA inside Kev

Small learned matrices adapt each selected projection




$$y = W_o v + (\alpha/r)BAv$$

$$\Delta W = (\alpha/r)BA$$

r = 16 α = 32 α/r = 2 LoRA dropout = 0.05

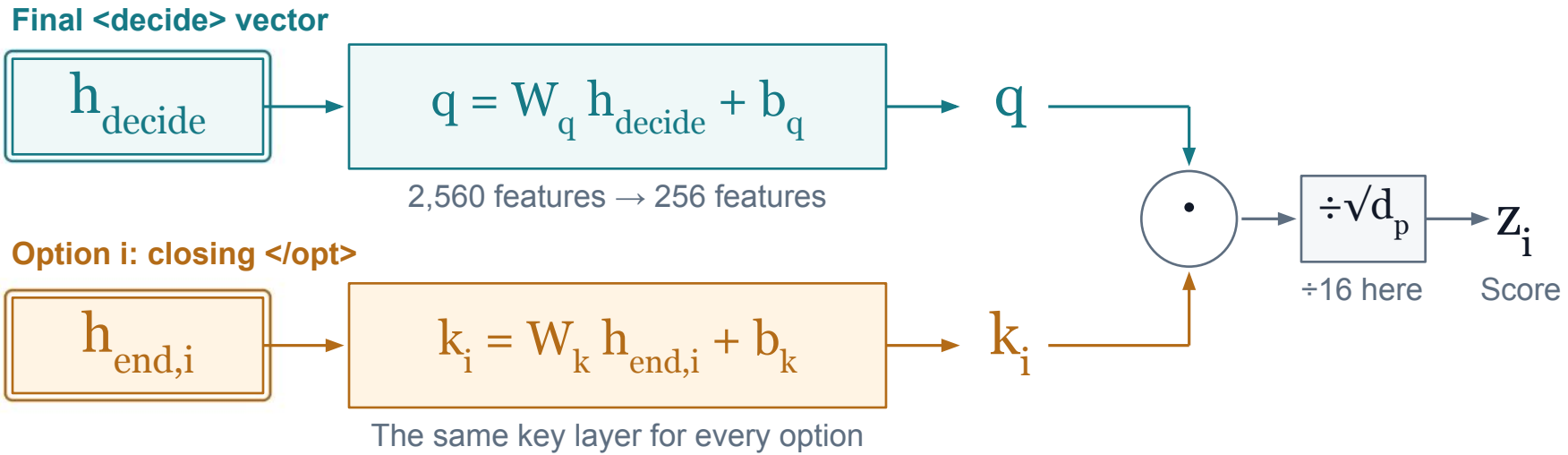
33.8M trainable parameters across LoRA and the pointer head

v: layer input y: layer output $d_{in}$ / $d_{out}$: input / output widths

One adapted projection. Kev uses all targets. Open animation controls.

# The pointer head scores every option

Two learned linear projections, then a scaled dot product




$$z_i = q^T k_i / \sqrt{d_p}$$

K legal options → K raw scores

$d = 2{,}560$ $d_p = 256$ $\sqrt{d_p} = 16$ Both projection layers train

Exact Kev pointer head. i indexes one option. Open animation controls.

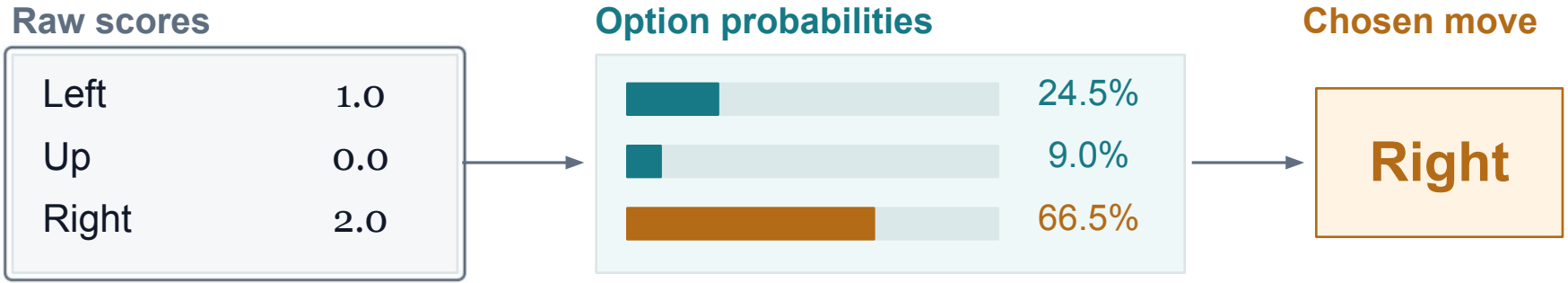
# Softmax and action selection

Softmax gives option probabilities. Argmax chooses the largest.

$$p(a_i \mid s) = \frac{\exp(z_i / T)}{\sum_{j=1}^{K} \exp(z_j / T)}$$

$T = 1$

Our fresh lab checkpoint



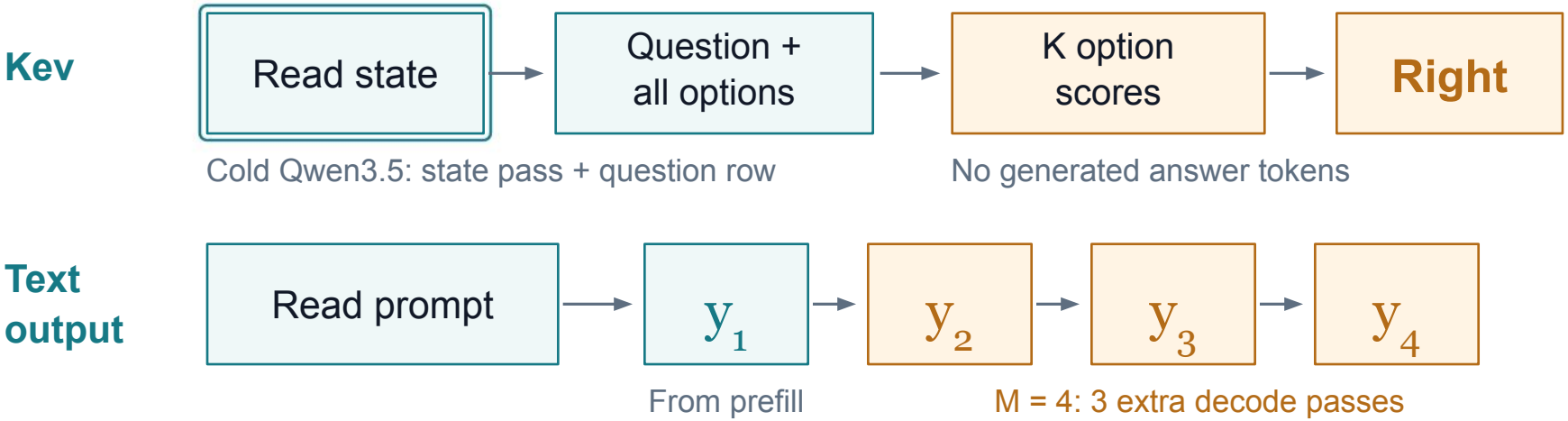

$$i^* = \operatorname{argmax}_i p(a_i \mid s)$$

$$a^* = a_{i^*}$$

Illustrative scores, not a live prediction. Open animation controls.

# Why Kev avoids the answer-token loop

The backbone reads the input. The pointer head returns K scores.




$$R_{Kev} \approx (K + 1)\, d\, d_p + K\, d_p$$

$$R_{text} \approx M\, d\, V$$

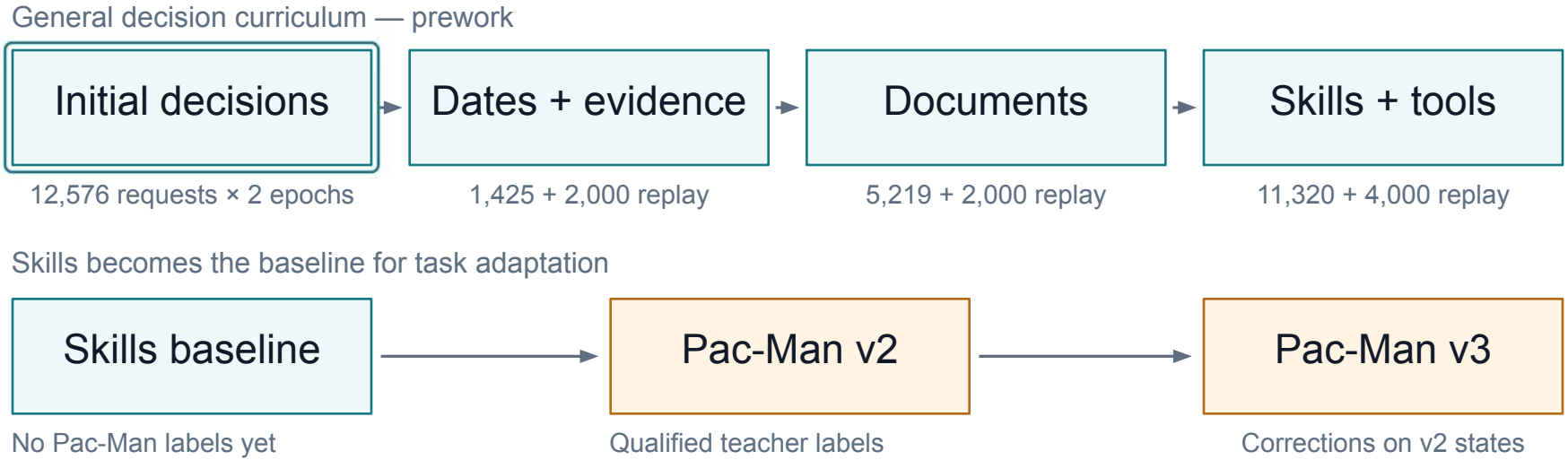
Readout multiplications

K choices, V vocabulary

Illustrative M = 4 answer. Counts exclude backbone work. Open animation controls.

# A curriculum, then task checkpoints

Each new stage loads its parent's LoRA and pointer head.




$$\varphi_{\text{child}}^{\,o} = \varphi_{\text{parent}} \qquad D_{\text{train}} = D_{\text{new}} \uplus D_{\text{replay}}$$

New stage: fresh optimizer. Parent checkpoint stays saved.

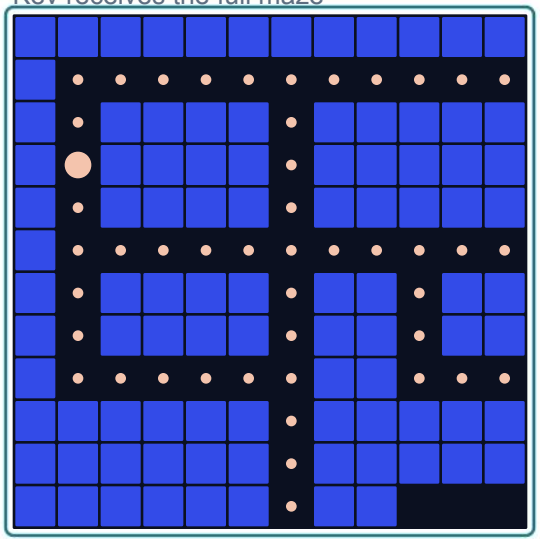
Same fixed Qwen base. Counts name requests; labels are not game wins. Open animation controls.

# The game state Kev actually receives

A structured board, moving actors, native timers and memory.

Display crop only

Kev receives the full maze



. pellet · o power pellet

## Player + four distinct ghosts

**Blinky** **Pinky** **Inky** **Clyde**

row/column · heading · pixel_offset · speed_phase

Ghosts: mode · frightened · next/pending turns

## Native clocks

timing.power.remaining_frames

chase/scatter · release · Elroy · fruit · eating pauses

## Movement and pellet history

recent_positions (up to 64) · destination_visits

decisions_since_last_pellet · visits_to_current_tile

$s_t = O(g_t)$

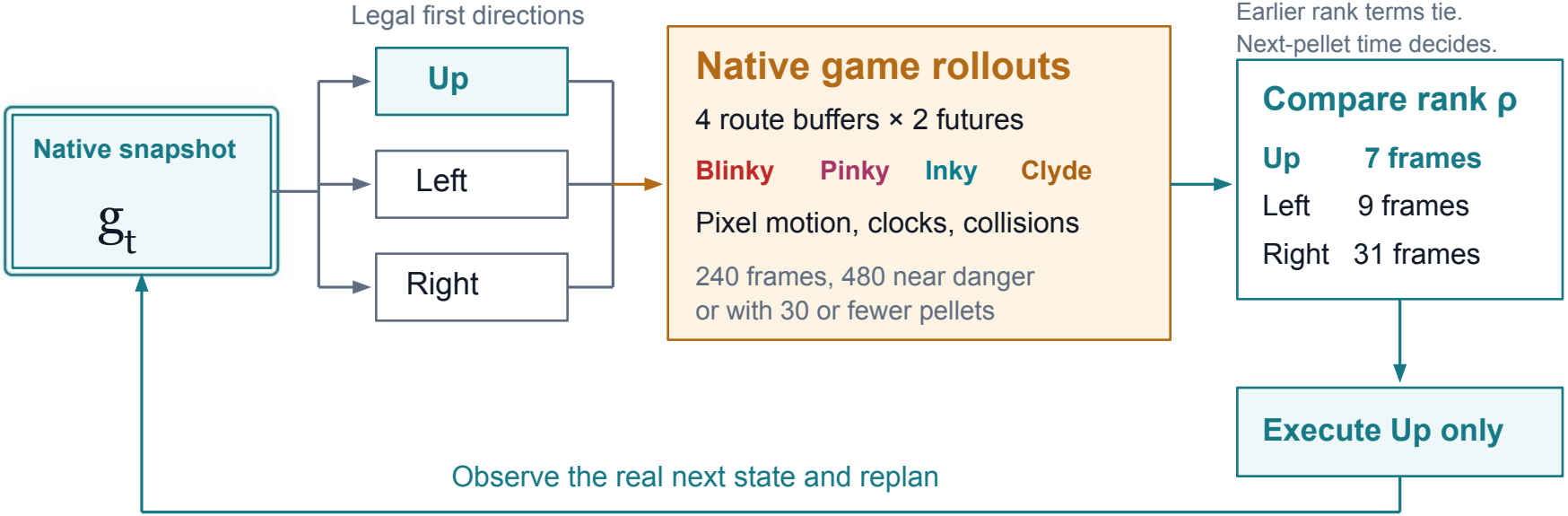
$a_t \in A_t$

Wall-legal directions

Teacher forecasts and labels stay outside this input. Open animation controls.

# Native rollout MPC teacher

Compare forecasts, execute one direction, then replan.




$$(a_t^T, b_t^*) = \operatorname{argmin}^{lex} \rho(g_t, a, b)$$

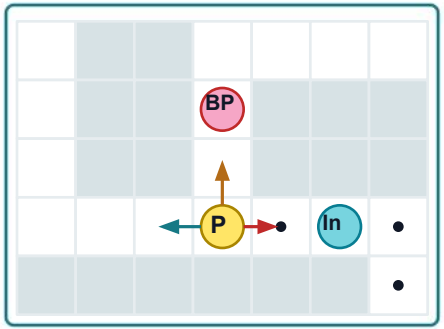
$a \in A_t$
$b \in \{1, 2, 4, 6\}$

Fresh native example. Finite policy portfolio. Open animation controls.

# Teacher route planning and future ranking

One native state exposes the trade-off between ghosts, loops and pellets.

**Seed 91009, level 1, turn 172**




BP: Blinky/Pinky. In: Inky. P: Pac-Man.

Display crop only. Full native game.

| First move | Death now | Dry cycle | Next pellet | Alive futures |
|---|---|---|---|---|
| **Left, b=6** | **0** | **0** | **137 frames** | **2 / 2, 484 frames** |
| Up, b=1 | 0 | 1 | none | 0 / 2, 9 frames |
| Right, b=1 | 1 | 0 | 6 frames | 0 / 2, 6 frames |

$$\rho = (E, C, O, D, -S, T, K, J, -W, -F, U, \bar{M})$$

**First unequal component: Left beats Right at E and Up at C.**

Food-route cost is one component of the inner rollout heuristic

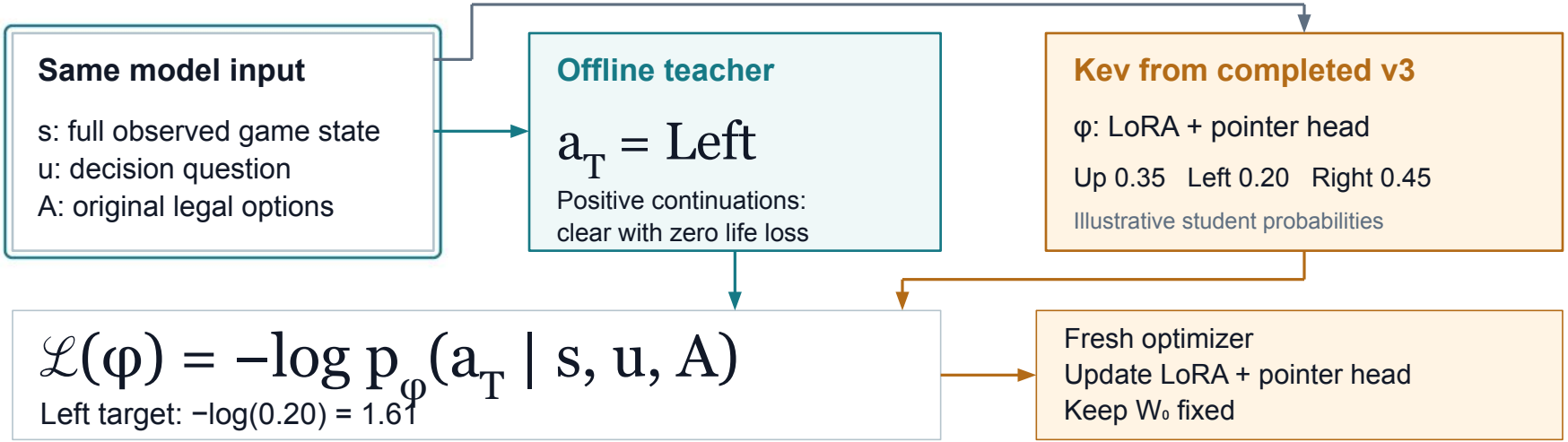
$$J_b(v) = \min_{p \in P} \{d(v,p) + 2\sum_{g \in G} [b - d(g,p)]_+\} + 2N(v)$$

**Execute Left only, then replan from the actual next state.**

Recorded CPU teacher forecasts. Finite policy portfolio. Open animation controls.

# Teacher decisions and Kev distillation

The teacher searches offline. Kev learns its chosen action.




| Targeted roots | Nearby decisions | Expert replay | Generic replay | Total requests |
|---|---|---|---|---|
| 3,660 | 1,220 | 912 | 304 | 6,096 |

11 roots are recorded v3 disagreements. Other hard cases use simulation.

Probability example only. Seed 91009 stays held out. Open animation controls.

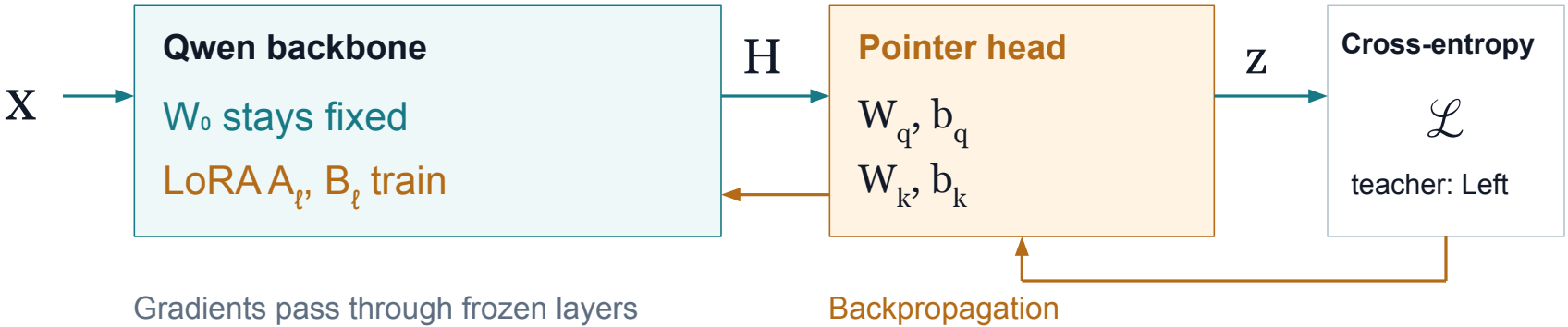
# How gradients update Kev

The loss flows backward. LoRA and the pointer head learn.

$$\partial \mathcal{L} / \partial z_i = p_i - t_i$$

Up +0.35  Left −0.80  Right +0.45

$t_i = 1$ for the teacher action, 0 otherwise. Values are illustrative.



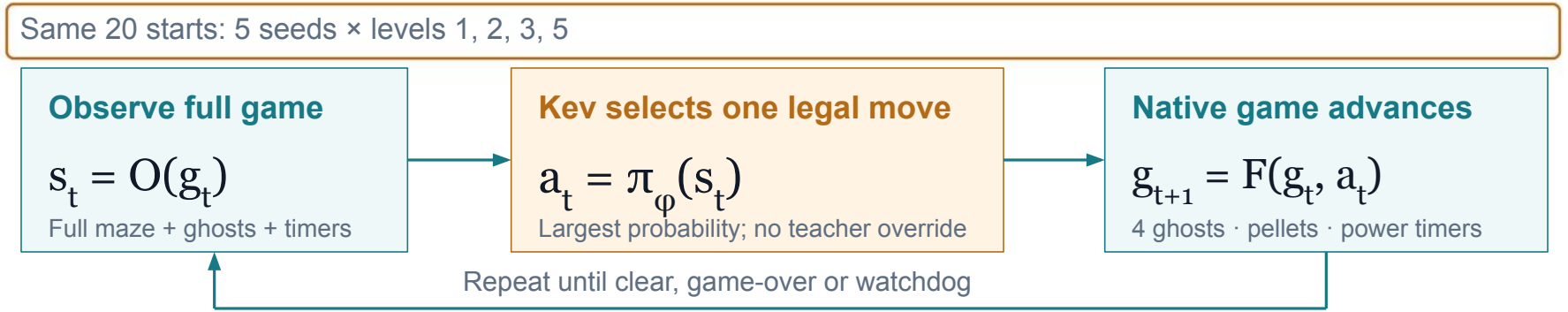

$$\varphi = \{A_\ell, B_\ell, W_q, b_q, W_k, b_k\}$$

Batch 4 × accum 2  Clip norm 1.0  Fused AdamW  peak lr 2e−5

Single-choice gradient. Frozen base. Open animation controls.

# Does the learned policy survive?

Run complete native games. Diagnose mistakes in saved traces.




$$C = (1/N) \sum_j 1[\text{clear}_j]$$

N = games; count genuine level clears only

$$E = 1000\ P / n_f$$

P: pellets; $n_f$: active native frames

Inspect deaths, dry spells and loops.

| Metric | Native-v3 | Teacher |
|---|---|---|
| Clears | 0/20 | 20/20 |
| Avoidable deaths | 22 | 0 |
| Pellets / 1k frames | 34.60 | 74.09 |

Recorded v3 / teacher. V4 must be measured.

Reference suite, not blind test. Open rollout controls.